



PAST

Project

Secondary Flight Storage

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Onboarding Logbook

Week 1 -

18th of March 2026

Tasks Completed :

- Selected onboarding project
- Created logbook design

Goals for Next Time:

- Learn basics of Altium designer
- Select a microcontroller
- Select a storage technology/module

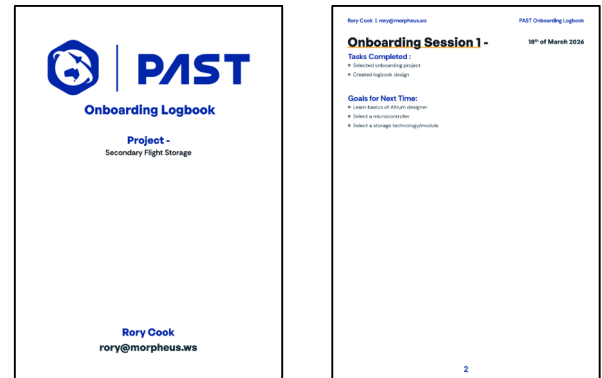


Figure 1: Logbook Design

Week 2 -

25th of March 2026

Tasks Completed :

- Created git repo for my work (link on cover page)
- Began creating research document
- Defined requirements
- Researched different flash storage topologies (NAND & NOR)
- Selected an IC to be used for the project (Infineon S25HL512TDPNH1010)

Chosen for its simple compact footprint and fast QSPI interface. It also has an easy to route design and far fewer redundant pins as compared to other available packages. It's NOR flash architecture also allowed for great durability and a faster read and write of the smaller data segments needed in an environment such as this one. It also uses SPI which makes the number of IOs used way smaller as compared to parallel which can take up to 3 or 4 times the number of IOs

- Chose a microcontroller (ESP32-S3-WROOM-1U)

I chose the ESP32-S3-WROOM-1U because of its ease of use and good IO options such as a built in USB to serial controller and easy to access SPI lanes, it also has an easy to route footprint and compact footprint with very few external components needed.

Problems:

- Initial challenges with spi flash chip due to a very large datasheet making it hard to work out if it was appropriate.
- Initially chose the wroom-1 which has a built in antenna which made certain parts hard to place so swapped to the 1U which just has an antenna connector.

Goals for Next Time:

- Read important info from datasheets
- Begin PCB design
- Learn how to use git with altium for version control

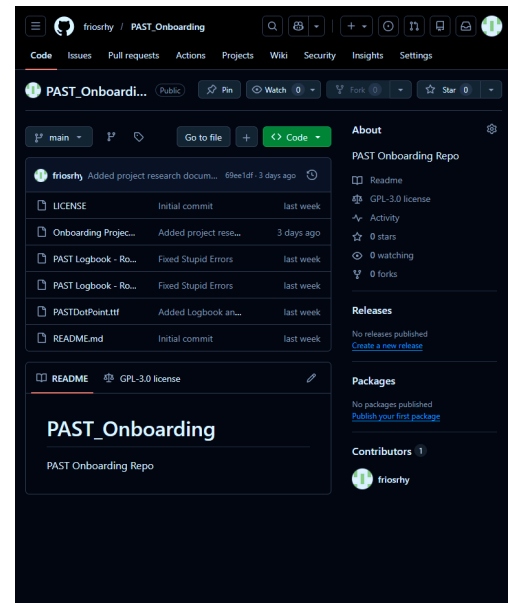


Figure 2: Github Repo

Week 3 -

1st of April 2026

Tasks Completed :

- Finished initial schematic of project as shown right. I essentially designed the whole thing so it serves dual purpose as a microcontroller so when it isn't being used for PAST anymore it retains its usefulness and can be used for other projects. It also uses minimal components aside from the microcontroller module so it is much easier to lay out than other designs I've made in the past.
- Started research on PCB layout best practices and how to ensure signal integrity.
- Most of the design choices for this board were made with simplicity in mind, first of which was the chosen usb-c receptacle, the UJ20-C-H-G-SMT-1-P16-TR, it was chosen because of its ease of soldering, ease of procurement given it has massive amounts of stock available and it has a fairly simple layout and it matches the USB 2.0 capability of the microcontroller.
- I chose the AZ1117-3.3CH2 as it is extremely common, very compact, can handle fairly high currents, more than enough to meet this board's needs, and has an ideal layout for the board design I intend on making.
- The chosen switches, the B3U-1000P from omron, are widely used, easy to solder, compact, can handle the necessary currents and are just one of the most common and easiest to work with switches on the market.

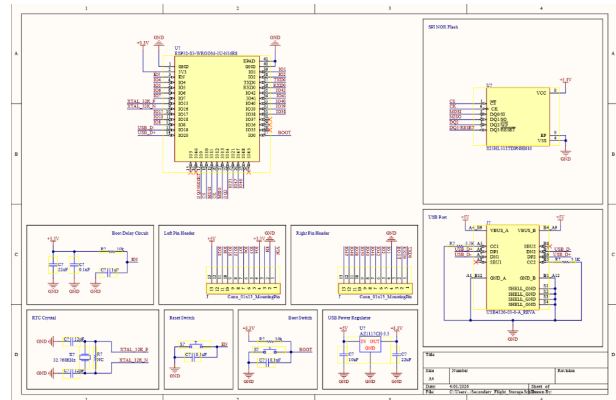


Figure 2: Initial schematic

Problems:

- Incorrect use of power flags with incorrect orientations
- Bad symbol design I made (has pins on bottom of symbol.)

Goals for Next Time:

- Make changes to schematic based on feedback
- Start learning how to do SPICE and electrical checks in altium
- Start to design PCB

Week 4 -

8th of April 2026

Tasks Completed :

- Rearchitected schematic using recommendations and feedback on my first design, I made sure that all my symbols only had pins on the left and right and tried to ensure that all my ground points were pointing down and my power symbols were facing up.
- Started using intlids properly and having footprints associated with all of my symbols, this let me make much better progress.
- Completed preliminary PCB layout in KiCAD to get a basic idea of layout and confirm ideas.

KiCAD was used due to my greater level of experience in the software and my ease of use compared to altium. It was also used as I could rough out a design much quicker giving me a basis in altium to work from

Problems:

- Couldn't work out how to properly import certain ecad files for parts in different format.
- Had to make a new pin header symbol and footprint which did not work for a bit until I reshuffled a few bits and pieces to get it working.

Goals for Next Time:

- Begin PCB layout
- Start to compile resources
- Start to design case

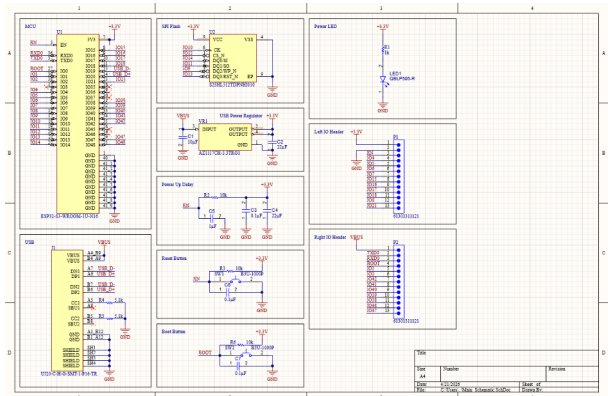


Figure 3: Rearchitected schematic

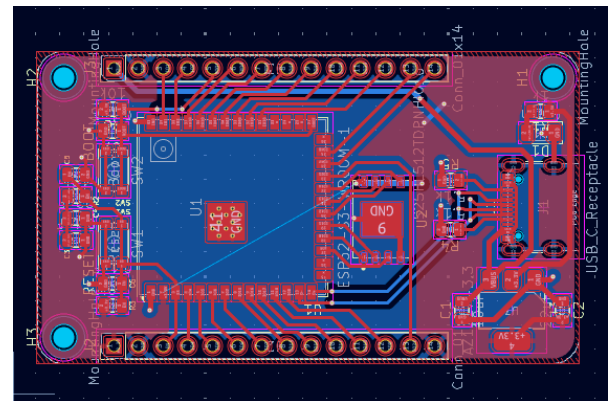


Figure 4: Basic PCB Layout In KiCAD

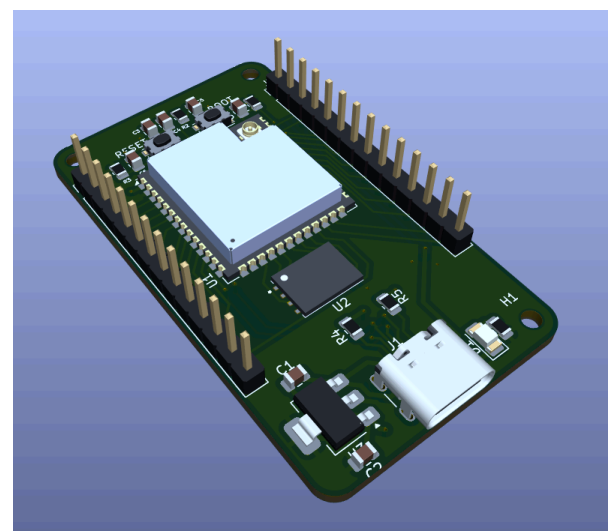


Figure 5: 3D Model of PCB

Week 5 -

22th of April 2026

Tasks Completed :

- Made basic PCB design and board outline in altium by importing board edge design as a dxf from solidworks and then bringing in footprints and trying to organise the silkscreen markings
- Added 3d models to some footprints
- Researched trace routing best practices for USB and SPI data lines. Subsequently learning that for USB you should try and route them as a pair and keep them running over an unbroken section of ground plane. For SPI I just aimed to keep them as short as possible and just like with the USB to try and keep them over an unbroken ground pour.
- I had design rule problems where manufacturer recommended footprints would violate DRC due to pad spacing and via positions and size

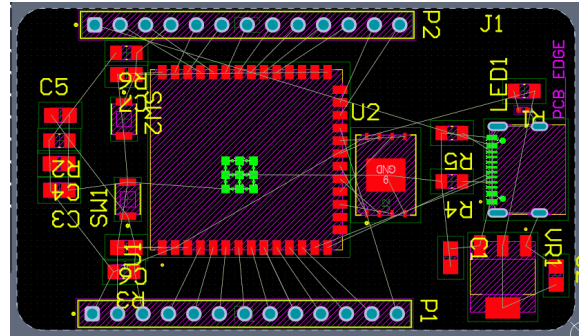


Figure 6: PCB Design in Altium

Problems:

- I could not work out how to move components precisely and position them relative to each other properly meaning that they aren't well aligned and the design is a bit messy looking.
- DRC errors when trying to route traces from usb which are stopping me from placing traces properly, despite using the manufacturer specified footprint.

Goals for Next Time:

- Make final PCB layout
- Get feedback on design
- Make 3D Model and start case design

Week 6 -

29nd of April 2026

Tasks Completed :

- Completed initial board design draft (still having teething issues with altium like pours and other such things)
- Got feedback on my logbook (needs more details and info)
- Got feedback on my PCB design (just need to focus on ensuring my pads on either side of footprints have the same no. of traces to minimise components flipping during soldering)
- Learned about altium's DRC quirks like shouting at you until you repour.
- Chose New resistors and capacitors, I went with 0805 as the larger size makes them much easier to hand solder and they are also more tolerant of larger current draws and voltages
- Swapped to a different AZ1117-3.3CH2 with a top ground pad for simpler routing of footprint.
- Learned more about how to do vias to ground plane for components such as avoiding placing vias within pads.
- Improved logbook and added links to my github page for the project
- Finished routing traces for my PCB design

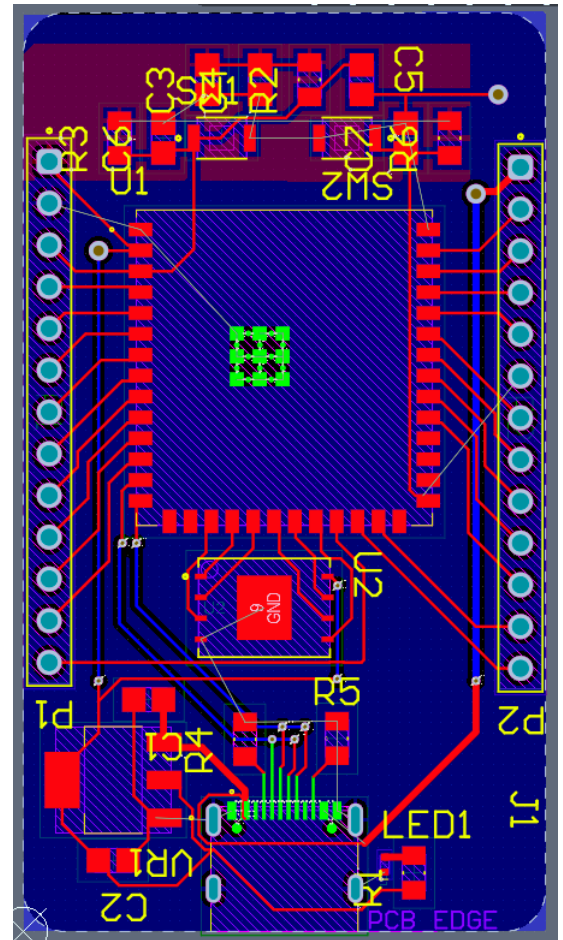


Figure 7: PCB Board Layout in Altium

Problems:

- Not able to get the ground plane to correctly conform to the shape of the board edge with edge clearance and such, leading to unusable gerbers if I was to manufacture the design.
- Had to run traces under the USB traces which is not recommended to ensure signal integrity and also can cause issues with losing connection in extreme cases.

Goals for Next Time:

- Improve logbook detail with data from my design documents and add github link
- Try to improve on design of logbooks
- Make 3D Model and start case
- Get somebody at work to check my pcb design
- Learn more about ground plane and how to enforce edge clearance

Week 7 -

5th of May 2026

Tasks Completed :

- Very Busy with study this week so didn't get much work done this week.

Goals for Next Time:

- Add screw/mounting holes to case
- Continue to design 3d cad model for pcb case
- Get final section of logbook ready for submission.

Week 8 -

13th of May 2026

Tasks Completed :

- Designed Case for PCB, bottom is cut off for usb-c cables of all shapes and sizes as the cutout sizes needed would be silly and wouldn't make sense. I also have two holes for accessing the boot and reset switches, then three holes in all but one corner with screw holes for m2 screws. I also chose to expose the pins so they can be used as pluggable headers for controlling other peripherals.
- Redid a few things on the schematic and added decoupling capacitors following feedback from PAST members. I also added an RGB LED for status indication and laid out the pcb more precisely to ensure everything lines up with CAD nicely. I also changed the layout to minimise the amount of traces on the back plane.
- Correctly imported design rules based on JLCPCB manufacturing capabilities.

Problems:

- buttons are not super accessible without a paperclip or similar
- Board edge is unprotected at one end
- LED is not very visible
- I didn't work out how to properly set the usb lines as differential pairs so I had to manually ensure clearance and length were roughly the same.

Evaluation:

Overall, I feel I did quite well on the overall design process and got fairly well accustomed to altium (though I still don't quite understand the library management). The board layout seems quite robust and passes the design checks that I put on it, though software would need to come next which I think I may struggle with. I also think that the process taught me a lot more about microcontrollers than I knew before and in addition the feedback I got really helped with my own designs and design skills in general.

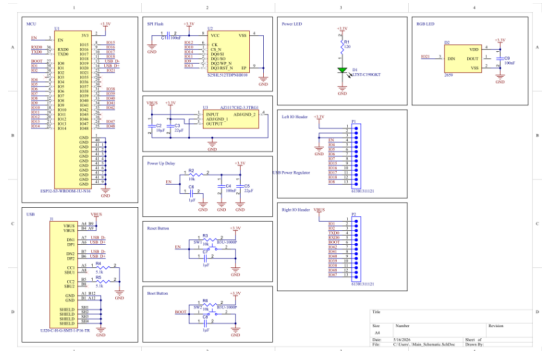


Figure 8: Updated schematic sheet

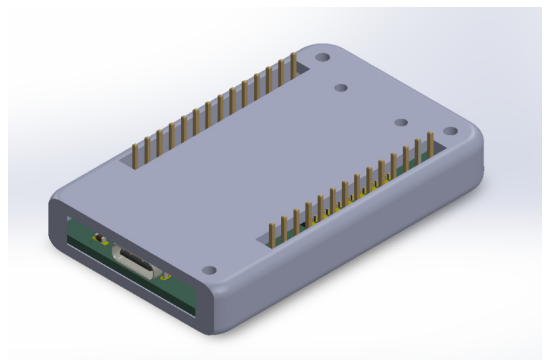


Figure 8: PCB Case Design

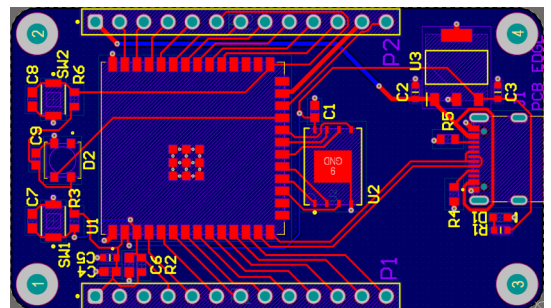


Figure 9: Finalised PCB Layout